

# Unemployment Analysis with Python

## Data Collection & Loading

This section focuses on importing the required Python libraries and loading the unemployment datasets for analysis.

```
In [1]: # Task 2: Unemployment Analysis with Python
# Cell 1: Data Collection & Loading - Importing Libraries and Loading Unemployme

import pandas as pd
import numpy as np
import matplotlib.pyplot as plt
import seaborn as sns

sns.set(style="whitegrid")

# Load the datasets
unemployment_df = pd.read_csv("Unemployment in India.csv")
unemployment_rate_df = pd.read_csv("Unemployment_Rate_upto_11_2020.csv")

print("Unemployment datasets loaded successfully!")

print("\nUnemployment in India Dataset:")
display(unemployment_df.head())

print("\nUnemployment Rate Dataset:")
display(unemployment_rate_df.head())
```

Unemployment datasets loaded successfully!

Unemployment in India Dataset:

|   | Region         | Date       | Frequency | Estimated<br>Unemployment<br>Rate (%) | Estimated<br>Employed | Estimated<br>Labour<br>Participation<br>Rate (%) | Area  |
|---|----------------|------------|-----------|---------------------------------------|-----------------------|--------------------------------------------------|-------|
| 0 | Andhra Pradesh | 31-05-2019 | Monthly   | 3.65                                  | 11999139.0            | 43.24                                            | Rural |
| 1 | Andhra Pradesh | 30-06-2019 | Monthly   | 3.05                                  | 11755881.0            | 42.05                                            | Rural |
| 2 | Andhra Pradesh | 31-07-2019 | Monthly   | 3.75                                  | 12086707.0            | 43.50                                            | Rural |
| 3 | Andhra Pradesh | 31-08-2019 | Monthly   | 3.32                                  | 12285693.0            | 43.97                                            | Rural |
| 4 | Andhra Pradesh | 30-09-2019 | Monthly   | 5.17                                  | 12256762.0            | 44.68                                            | Rural |

Unemployment Rate Dataset:

|   | Region         | Date       | Frequency | Estimated<br>Unemployment<br>Rate (%) | Estimated<br>Employed | Estimated<br>Labour<br>Participation<br>Rate (%) | Region.1 | longitud |
|---|----------------|------------|-----------|---------------------------------------|-----------------------|--------------------------------------------------|----------|----------|
| 0 | Andhra Pradesh | 31-01-2020 | M         | 5.48                                  | 16635535              | 41.02                                            | South    | 15.91    |
| 1 | Andhra Pradesh | 29-02-2020 | M         | 5.83                                  | 16545652              | 40.90                                            | South    | 15.91    |
| 2 | Andhra Pradesh | 31-03-2020 | M         | 5.79                                  | 15881197              | 39.18                                            | South    | 15.91    |
| 3 | Andhra Pradesh | 30-04-2020 | M         | 20.51                                 | 11336911              | 33.10                                            | South    | 15.91    |
| 4 | Andhra Pradesh | 31-05-2020 | M         | 17.43                                 | 12988845              | 36.46                                            | South    | 15.91    |



## Data Understanding & Exploration

This section examines the structure, data types, missing values and statistical characteristics of the unemployment datasets.

In [2]: *# Cell 2: Data Understanding & Exploration - Inspecting Dataset Structure, Data*

```
print("==== Unemployment in India Dataset ====")

print("\nShape:")
print(unemployment_df.shape)

print("\nColumns:")
print(unemployment_df.columns.tolist())

print("\nData Types:")
print(unemployment_df.dtypes)

print("\nMissing Values:")
print(unemployment_df.isnull().sum())

print("\nDescriptive Statistics:")
display(unemployment_df.describe(include="all"))

print("\n\n==== Unemployment Rate Dataset ====")

print("\nShape:")
print(unemployment_rate_df.shape)

print("\nColumns:")
print(unemployment_rate_df.columns.tolist())

print("\nData Types:")
print(unemployment_rate_df.dtypes)

print("\nMissing Values:")
print(unemployment_rate_df.isnull().sum())

print("\nDescriptive Statistics:")
display(unemployment_rate_df.describe(include="all"))
```

# ===== Unemployment in India Dataset =====

Shape:

(768, 7)

Columns:

['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate (%)', ' Estimated Employed', ' Estimated Labour Participation Rate (%)', 'Area']

Data Types:

```
Region          str
Date            str
Frequency       str
Estimated Unemployment Rate (%) float64
Estimated Employed float64
Estimated Labour Participation Rate (%) float64
Area            str
dtype: object
```

Missing Values:

```
Region          28
Date            28
Frequency       28
Estimated Unemployment Rate (%) 28
Estimated Employed 28
Estimated Labour Participation Rate (%) 28
Area            28
dtype: int64
```

Descriptive Statistics:

|               | Region         | Date       | Frequency | Estimated<br>Unemployment<br>Rate (%) | Estimated<br>Employed | Estimated<br>Labour<br>Participation<br>Rate (%) | Area  |
|---------------|----------------|------------|-----------|---------------------------------------|-----------------------|--------------------------------------------------|-------|
| <b>count</b>  | 740            | 740        | 740       | 740.000000                            | 7.400000e+02          | 740.000000                                       | 740   |
| <b>unique</b> | 28             | 14         | 2         | NaN                                   | NaN                   | NaN                                              | 2     |
| <b>top</b>    | Andhra Pradesh | 31-10-2019 | Monthly   | NaN                                   | NaN                   | NaN                                              | Urban |
| <b>freq</b>   | 28             | 55         | 381       | NaN                                   | NaN                   | NaN                                              | 381   |
| <b>mean</b>   | NaN            | NaN        | NaN       | 11.787946                             | 7.204460e+06          | 42.630122                                        | NaN   |
| <b>std</b>    | NaN            | NaN        | NaN       | 10.721298                             | 8.087988e+06          | 8.111094                                         | NaN   |
| <b>min</b>    | NaN            | NaN        | NaN       | 0.000000                              | 4.942000e+04          | 13.330000                                        | NaN   |
| <b>25%</b>    | NaN            | NaN        | NaN       | 4.657500                              | 1.190404e+06          | 38.062500                                        | NaN   |
| <b>50%</b>    | NaN            | NaN        | NaN       | 8.350000                              | 4.744178e+06          | 41.160000                                        | NaN   |
| <b>75%</b>    | NaN            | NaN        | NaN       | 15.887500                             | 1.127549e+07          | 45.505000                                        | NaN   |
| <b>max</b>    | NaN            | NaN        | NaN       | 76.740000                             | 4.577751e+07          | 72.570000                                        | NaN   |

===== Unemployment Rate Dataset =====

Shape:

(267, 9)

Columns:

['Region', ' Date', ' Frequency', ' Estimated Unemployment Rate (%)', ' Estimated  
Employed', ' Estimated Labour Participation Rate (%)', 'Region.1', 'longitude',  
'latitude']

Data Types:

|                                         |         |
|-----------------------------------------|---------|
| Region                                  | str     |
| Date                                    | str     |
| Frequency                               | str     |
| Estimated Unemployment Rate (%)         | float64 |
| Estimated Employed                      | int64   |
| Estimated Labour Participation Rate (%) | float64 |
| Region.1                                | str     |
| longitude                               | float64 |
| latitude                                | float64 |
| dtype:                                  | object  |

Missing Values:

|                                         |       |
|-----------------------------------------|-------|
| Region                                  | 0     |
| Date                                    | 0     |
| Frequency                               | 0     |
| Estimated Unemployment Rate (%)         | 0     |
| Estimated Employed                      | 0     |
| Estimated Labour Participation Rate (%) | 0     |
| Region.1                                | 0     |
| longitude                               | 0     |
| latitude                                | 0     |
| dtype:                                  | int64 |

Descriptive Statistics:

|               | Region         | Date       | Frequency | Estimated<br>Unemployment<br>Rate (%) | Estimated<br>Employed | Estimated<br>Labour<br>Participation<br>Rate (%) | Region.1 |
|---------------|----------------|------------|-----------|---------------------------------------|-----------------------|--------------------------------------------------|----------|
| <b>count</b>  | 267            | 267        | 267       | 267.000000                            | 2.670000e+02          | 267.000000                                       | 267      |
| <b>unique</b> | 27             | 10         | 1         | NaN                                   | NaN                   | NaN                                              | 5        |
| <b>top</b>    | Andhra Pradesh | 31-03-2020 | M         | NaN                                   | NaN                   | NaN                                              | North    |
| <b>freq</b>   | 10             | 27         | 267       | NaN                                   | NaN                   | NaN                                              | 79       |
| <b>mean</b>   | NaN            | NaN        | NaN       | 12.236929                             | 1.396211e+07          | 41.681573                                        | NaN      |
| <b>std</b>    | NaN            | NaN        | NaN       | 10.803283                             | 1.336632e+07          | 7.845419                                         | NaN      |
| <b>min</b>    | NaN            | NaN        | NaN       | 0.500000                              | 1.175420e+05          | 16.770000                                        | NaN      |
| <b>25%</b>    | NaN            | NaN        | NaN       | 4.845000                              | 2.838930e+06          | 37.265000                                        | NaN      |
| <b>50%</b>    | NaN            | NaN        | NaN       | 9.650000                              | 9.732417e+06          | 40.390000                                        | NaN      |
| <b>75%</b>    | NaN            | NaN        | NaN       | 16.755000                             | 2.187869e+07          | 44.055000                                        | NaN      |
| <b>max</b>    | NaN            | NaN        | NaN       | 75.850000                             | 5.943376e+07          | 69.690000                                        | NaN      |

## Data Cleaning & Preparation

This section prepares the datasets for analysis by standardizing column names, converting dates, handling missing values and organizing the data chronologically.

```
In [4]: # Cell 3: Data Cleaning & Preparation - Standardizing Columns, Converting Dates

# Remove extra spaces from column names
unemployment_df.columns = unemployment_df.columns.str.strip()
unemployment_rate_df.columns = unemployment_rate_df.columns.str.strip()

# Display the column names after cleaning
print("Unemployment in India columns:")
print(unemployment_df.columns.tolist())

print("\nUnemployment Rate columns:")
print(unemployment_rate_df.columns.tolist())

# Convert Date columns into datetime format
unemployment_df["Date"] = pd.to_datetime(unemployment_df["Date"], errors="coerce")
unemployment_rate_df["Date"] = pd.to_datetime(unemployment_rate_df["Date"], errors="coerce")

# Remove rows containing missing values
unemployment_df = unemployment_df.dropna().copy()
unemployment_rate_df = unemployment_rate_df.dropna().copy()

# Sort data by date
unemployment_df = unemployment_df.sort_values("Date").reset_index(drop=True)
unemployment_rate_df = unemployment_rate_df.sort_values("Date").reset_index(drop=True)
```

```
print("\nData cleaning and preparation completed successfully!")

print("\nRemaining missing values:")
print("Unemployment in India:", unemployment_df.isnull().sum().sum())
print("Unemployment Rate:", unemployment_rate_df.isnull().sum().sum())
```

Unemployment in India columns:

```
['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)', 'Estimated Employed', 'Estimated Labour Participation Rate (%)', 'Area']
```

Unemployment Rate columns:

```
['Region', 'Date', 'Frequency', 'Estimated Unemployment Rate (%)', 'Estimated Employed', 'Estimated Labour Participation Rate (%)', 'Region.1', 'longitude', 'latitude']
```

Data cleaning and preparation completed successfully!

Remaining missing values:

Unemployment in India: 0

Unemployment Rate: 0

## Exploratory Data Analysis (EDA)

This section analyzes the distribution of unemployment rates and identifies the spread and possible outliers using visualizations.

In [5]: *# Cell 4: Exploratory Data Analysis - Analyzing Unemployment Rate Distribution a*

```
# Create histogram to understand the distribution of unemployment rates
plt.figure(figsize=(10, 6))
```

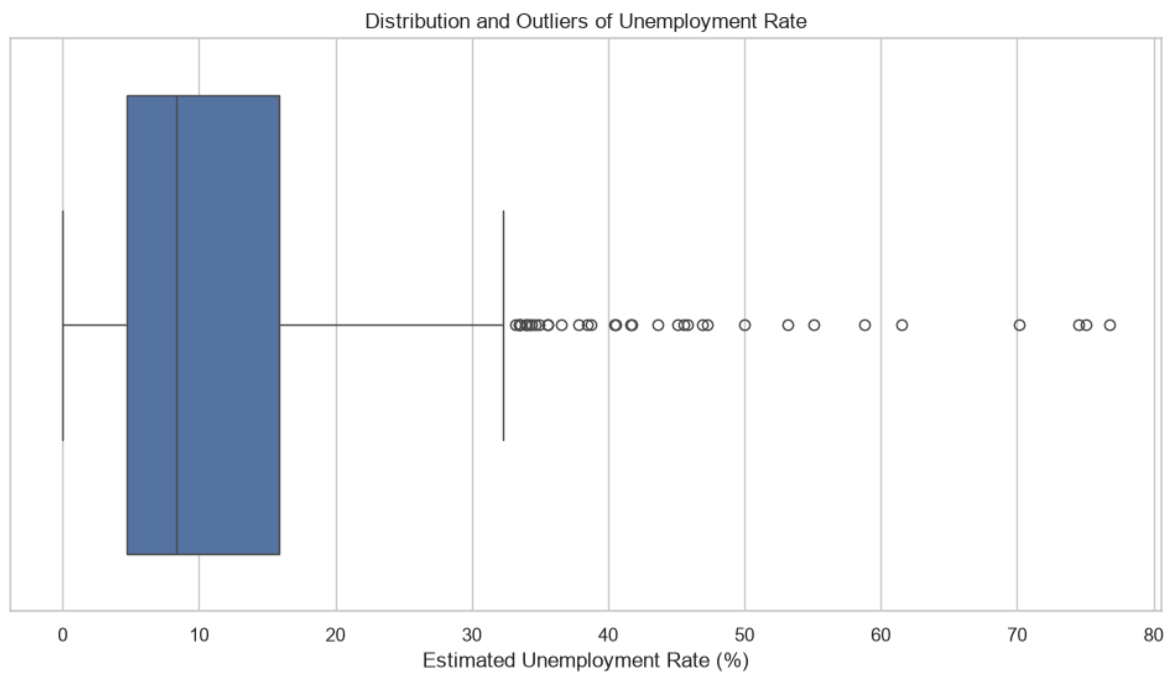
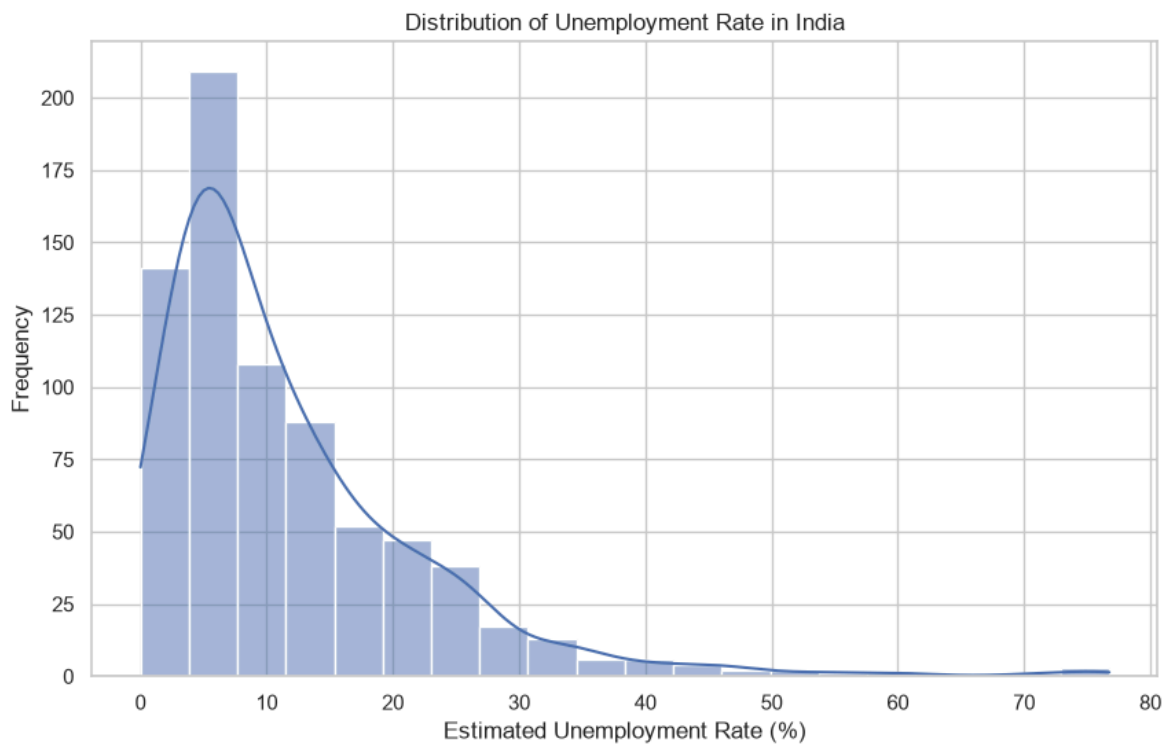
```
sns.histplot(
    unemployment_df["Estimated Unemployment Rate (%)"],
    bins=20,
    kde=True
)
```

```
plt.title("Distribution of Unemployment Rate in India")
plt.xlabel("Estimated Unemployment Rate (%)")
plt.ylabel("Frequency")
plt.show()
```

```
# Create boxplot to identify spread and possible outliers
plt.figure(figsize=(12, 6))
```

```
sns.boxplot(
    x=unemployment_df["Estimated Unemployment Rate (%)"]
)
```

```
plt.title("Distribution and Outliers of Unemployment Rate")
plt.xlabel("Estimated Unemployment Rate (%)")
plt.show()
```



## Regional Analysis

This section compares the average unemployment rates across different states and identifies regions with higher unemployment rates.

```
In [6]: # Cell 5: Regional Analysis - Comparing Unemployment Rates Across States

# Calculate average unemployment rate for each state
state_unemployment = (
    unemployment_df.groupby("Region")["Estimated Unemployment Rate (%)"]
    .mean()
    .sort_values(ascending=False)
)
```

```

# Display the average unemployment rate of each state
print("Average Unemployment Rate by State:")
display(state_unemployment)

# Select the top 10 states with the highest average unemployment rate
top_10_states = state_unemployment.head(10)

# Create a bar chart for the top 10 states
plt.figure(figsize=(12, 6))

sns.barplot(
    x=top_10_states.values,
    y=top_10_states.index
)

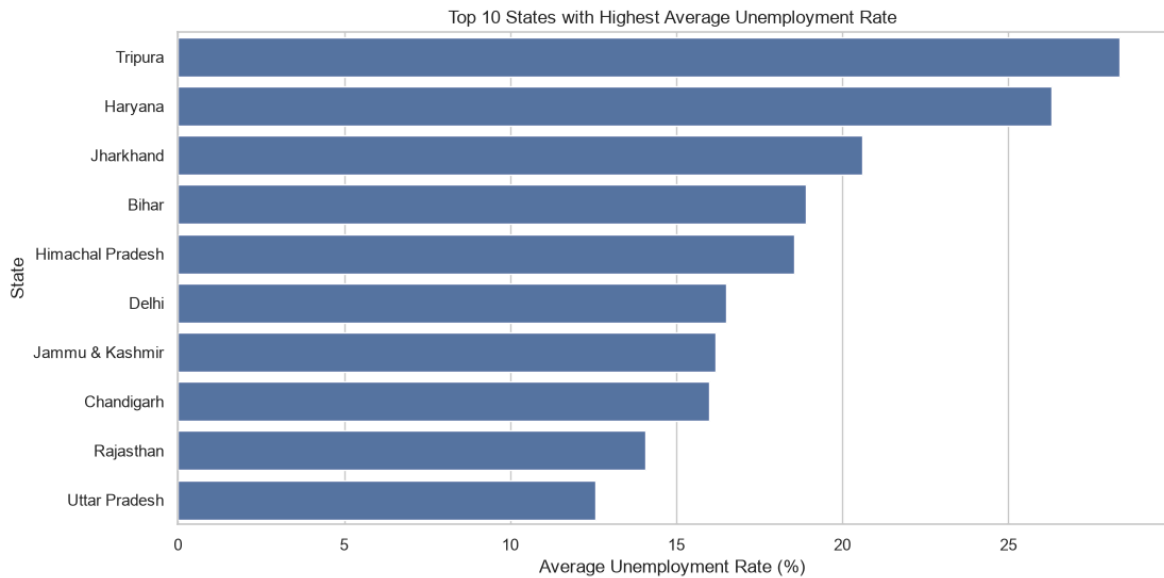
plt.title("Top 10 States with Highest Average Unemployment Rate")
plt.xlabel("Average Unemployment Rate (%)")
plt.ylabel("State")
plt.show()

```

Average Unemployment Rate by State:

| Region           |           |
|------------------|-----------|
| Tripura          | 28.350357 |
| Haryana          | 26.283214 |
| Jharkhand        | 20.585000 |
| Bihar            | 18.918214 |
| Himachal Pradesh | 18.540357 |
| Delhi            | 16.495357 |
| Jammu & Kashmir  | 16.188571 |
| Chandigarh       | 15.991667 |
| Rajasthan        | 14.058214 |
| Uttar Pradesh    | 12.551429 |
| Punjab           | 12.031071 |
| Puducherry       | 10.215000 |
| Kerala           | 10.123929 |
| Tamil Nadu       | 9.284286  |
| Goa              | 9.274167  |
| Chhattisgarh     | 9.240357  |
| West Bengal      | 8.124643  |
| Telangana        | 7.737857  |
| Maharashtra      | 7.557500  |
| Andhra Pradesh   | 7.477143  |
| Madhya Pradesh   | 7.406429  |
| Sikkim           | 7.249412  |
| Karnataka        | 6.676071  |
| Gujarat          | 6.663929  |
| Uttarakhand      | 6.582963  |
| Assam            | 6.428077  |
| Odisha           | 5.657857  |
| Meghalaya        | 4.798889  |

Name: Estimated Unemployment Rate (%), dtype: float64



## Time-Series Analysis

This section examines changes in the unemployment rate over time and visualizes the overall unemployment trend.

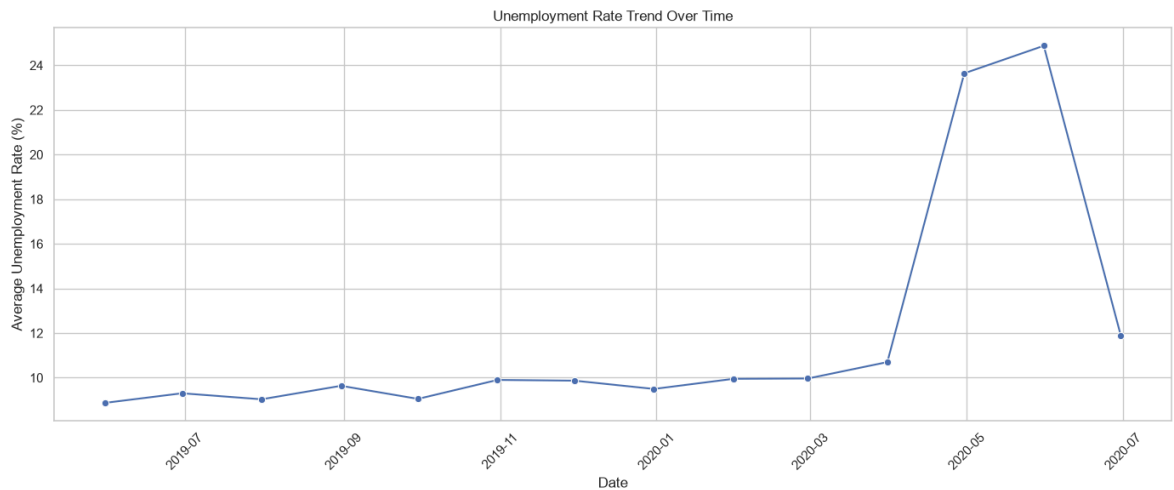
In [7]: *# Cell 6: Time-Series Analysis - Analyzing Unemployment Trends Over Time*

```
# Calculate the average unemployment rate for each date
monthly_unemployment = (
    unemployment_df.groupby("Date")["Estimated Unemployment Rate (%)"]
    .mean()
    .reset_index()
)

# Create a line chart to visualize unemployment trends over time
plt.figure(figsize=(14, 6))

sns.lineplot(
    data=monthly_unemployment,
    x="Date",
    y="Estimated Unemployment Rate (%)",
    marker="o"
)

plt.title("Unemployment Rate Trend Over Time")
plt.xlabel("Date")
plt.ylabel("Average Unemployment Rate (%)")
plt.xticks(rotation=45)
plt.tight_layout()
plt.show()
```



## Insights & Conclusion

This section summarizes the major findings from the unemployment analysis, including overall unemployment levels, regional differences and trends over time.

```
In [8]: # Cell 7: Insights & Conclusion - Identifying Major Unemployment Trends

# Calculate overall unemployment statistics
average_unemployment = unemployment_df["Estimated Unemployment Rate (%)"].mean()
maximum_unemployment = unemployment_df["Estimated Unemployment Rate (%)"].max()
minimum_unemployment = unemployment_df["Estimated Unemployment Rate (%)"].min()

# Find the state with the highest average unemployment rate
highest_state = state_unemployment.idxmax()
highest_state_rate = state_unemployment.max()

# Find the state with the lowest average unemployment rate
lowest_state = state_unemployment.idxmin()
lowest_state_rate = state_unemployment.min()

# Display the major findings
print("==== Unemployment Analysis - Key Insights ====")

print(f"\nOverall Average Unemployment Rate: {average_unemployment:.2f}%")

print(f"Highest Recorded Unemployment Rate: {maximum_unemployment:.2f}%")

print(f"Lowest Recorded Unemployment Rate: {minimum_unemployment:.2f}%")

print(f"\nState with Highest Average Unemployment Rate: {highest_state}")
print(f"Average Rate: {highest_state_rate:.2f}%")

print(f"\nState with Lowest Average Unemployment Rate: {lowest_state}")
print(f"Average Rate: {lowest_state_rate:.2f}%")

print("\nThe analysis shows variations in unemployment rates across different st
print("The visualizations help identify unemployment distribution, regional diff
```

===== Unemployment Analysis - Key Insights =====

Overall Average Unemployment Rate: 11.79%  
Highest Recorded Unemployment Rate: 76.74%  
Lowest Recorded Unemployment Rate: 0.00%

State with Highest Average Unemployment Rate: Tripura  
Average Rate: 28.35%

State with Lowest Average Unemployment Rate: Meghalaya  
Average Rate: 4.80%

The analysis shows variations in unemployment rates across different states and time periods.

The visualizations help identify unemployment distribution, regional differences and trends over time.

In [ ]:

